In [6]:
```
# Exp :- 12
```

In [7]:
```
# Aim : To perform Operation of KNN classifiers
```

In [8]:
```
#Name:Arya patil
#Roll:42
#sec:C
#subject:DSS
#Date:
```

In [9]:
```
import pandas as pd
import numpy as np
```

In [10…
```
import os
```

In [11…
```
os.getcwd()
```

Out[11]:
```
'C:\\Users\\aryap'
```

In [12…
```
os.chdir("C:\\Users\\aryap\\Downloads")
```

In [13…
```
data=pd.read_csv("Iris.csv")
```

In [14…
```
data.head()
```

Out[14]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

In [15…
```
data.tail()
```

Out[15]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

In [16…
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Id             150 non-null    int64  
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object 
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

In [17…
```
data.describe()
```

Out[17]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

In [18…
```
data.shape
```

Out[18]:
```
(150, 6)
```

In [19…
```
data.size
```

Out[19]:
```
900
```

In [20…
```
data.ndim
```

Out[20]:
```
2
```

In [21…
```
data.isna()
```

Out[21]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
145	False	False	False	False	False	False
146	False	False	False	False	False	False
147	False	False	False	False	False	False
148	False	False	False	False	False	False
149	False	False	False	False	False	False

150 rows × 6 columns

In [22…
```
data.isna().any()
```

Out[22]:
```
Id               False
SepalLengthCm    False
SepalWidthCm     False
PetalLengthCm    False
PetalWidthCm     False
Species          False
dtype: bool
```

In [23…
```
data.isna().sum()
```

Out[23]:
```
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

In [24…
```
x=data.drop("Species", axis=1)
y=data["Species"]
```

In [25…
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2 ,random_
```

In [26…
```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
```

In [27…
```
knn=KNeighborsClassifier()
```

In [28…
```
knn.fit(x_train, y_train)
```

Out[28]:
```
KNeighborsClassifier()
```

In [29…
```
y_pred2=knn.predict(x_test)
```

In [30…
```
accuracy = accuracy_score(y_test, y_pred2)
```

In [31…
```
accuracy
```

Out[31]:
```
1.0
```

In [ ]:

In [ ]:

In [ ]:

In [ ]: